



Bahir Dar University
"Wisdom at the Source of the Blue Nile"

DEPARTMENT OF SOFTWARE ENGINEERING

FACUALTY OF COMPUTING

OPERATING SYSTEM

IDIVIDIUAl ASSIGNMENT

SECTION-A

TITLE: BackBox linux Operating System

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Table of Contents

- 1. Introduction**
- 2. Features of BackBox Linux**
- 3. Objectives of the Project**
- 4. System Requirements**
 - 4.1 Hardware Requirements
 - 4.2 Software Requirements
- 5. Installation Process**
 - 5.1 Installing VMware Workstation
 - 5.2 Downloading BackBox Linux ISO
 - 5.3 Creating Virtual Machine
 - 5.4 Installing BackBox Linux
 - 5.5 Completing Installation
- 6. Issues (Problems Faced)**
- 7. Solutions**
- 8. Advantages**
- 9. Disadvantages**
- 10. Conclusion (BackBox Linux Installation)**
- 11. Virtualization in Modern Operating Systems**
 - 11.1 What is Virtualization
 - 11.2 Why Use Virtualization
 - 11.3 How Virtualization Works
 - 11.4 Types of Hypervisors
- 12. Final Conclusion**
- 13. References**

BackBox Linux OS

BackBox Linux is a Linux operating system based on Ubuntu. It is specially designed for **security testing and penetration testing**.

This system is made to be **fast and lightweight**, meaning it does not use too many computer resources. At the same time, it includes updated and useful open-source security tools.

BackBox is also a **free and open-source project**, which means anyone can use it and contribute to it. The goal of the project is to help people learn about cybersecurity and create awareness about system security.

One of the main goals of BackBox is to give **high performance while using very little system resources**. Because of this, it can even run on older computers without major problems.

In practice, BackBox comes with many tools used in ethical hacking, such as:

- Network scanning tools
- Vulnerability checking tools
- Exploitation tools
- Digital forensics tools

All these tools are organized and updated through its software repositories.

BackBox is developed and maintained by a **community**, not a company, so there is no official paid support. However, the system is regularly updated with new versions.

For example, **BackBox 9 (released in October 2024)** includes:

- Linux kernel 6.8
- Compatibility with Ubuntu 24.04 LTS

Objectives

The main purpose of this documentation is to:

- Show how to install BackBox Linux using virtual machines like **Oracle VirtualBox** and **VMware Workstation**
- Explain the installation process step by step
- Describe any problems faced during installation and how to solve them
- Analyze the filesystem support of BackBox
- Explain the advantages and disadvantages of using BackBox
- Understand how virtualization helps in setting up and running BackBox

The overall goal is to create a **safe environment for security testing** and also learn how virtualization works in real practice.

Requirements

Hardware Requirements

BackBox Linux does not need very powerful hardware. The basic requirements are:

- A **64-bit processor (CPU)**
- At least **1 GB RAM** (but 2 GB or more is better for smooth performance)
- At least **30 GB of disk space**
- Basic graphics support (around 800×600 resolution is enough)

For virtualization, the host computer must also:

- Support **hardware virtualization** (Intel VT-x or AMD-V)
- Have enough extra RAM and storage to run the virtual machine properly

Software Requirements

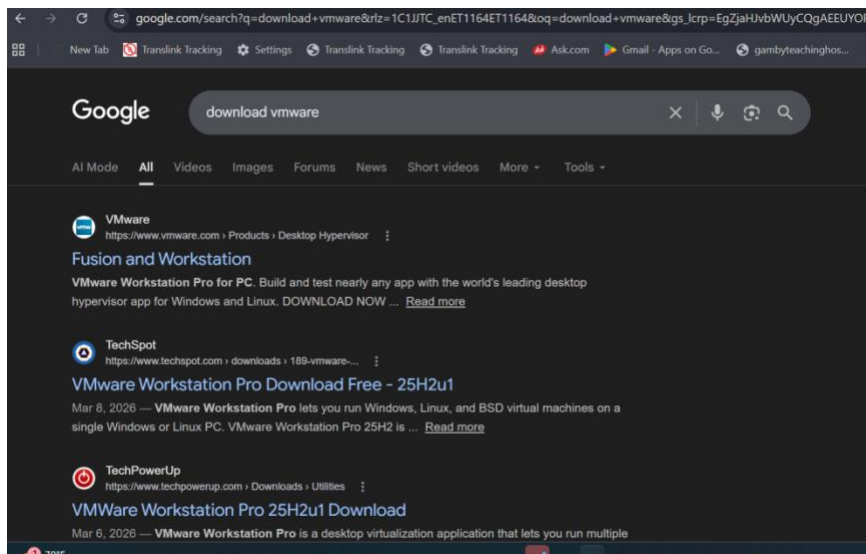
To install and run BackBox Linux, you need:

- The **BackBox ISO image file**
- A virtualization software such as:
 - **Oracle VirtualBox**
 - **VMware Workstation**

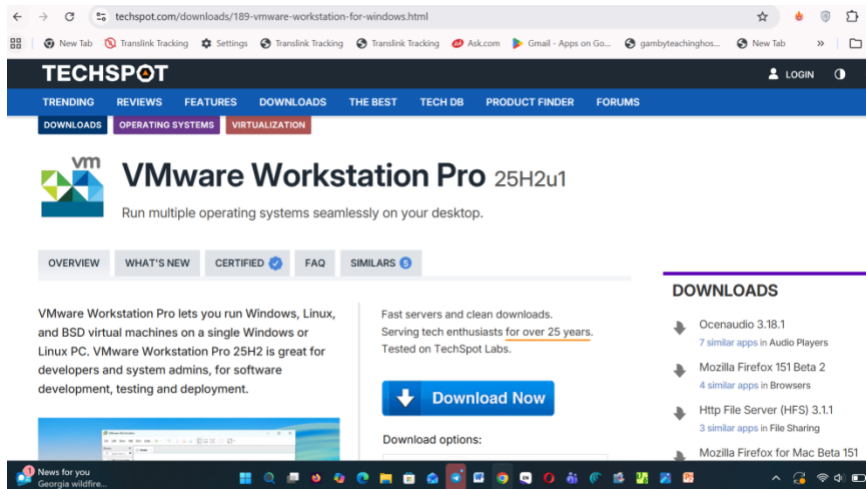
These tools allow you to create and run virtual machines on your computer.

Installation Steps

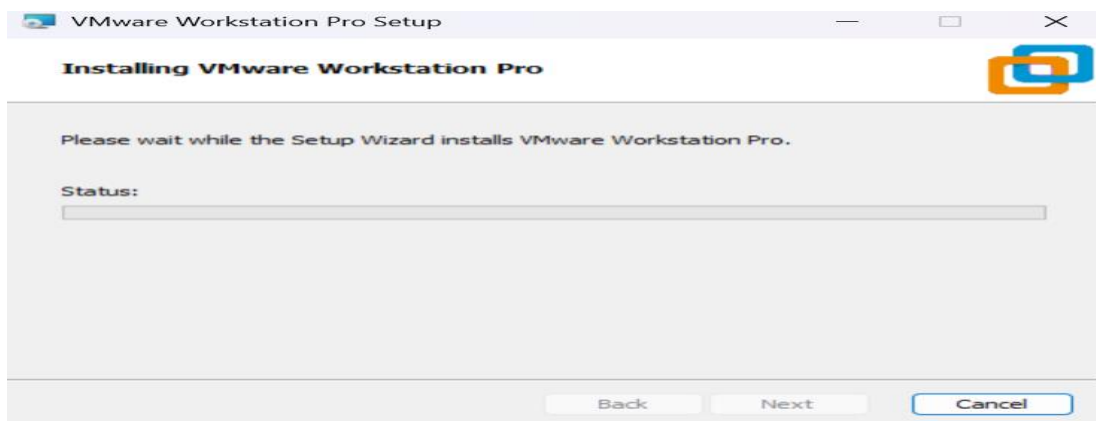
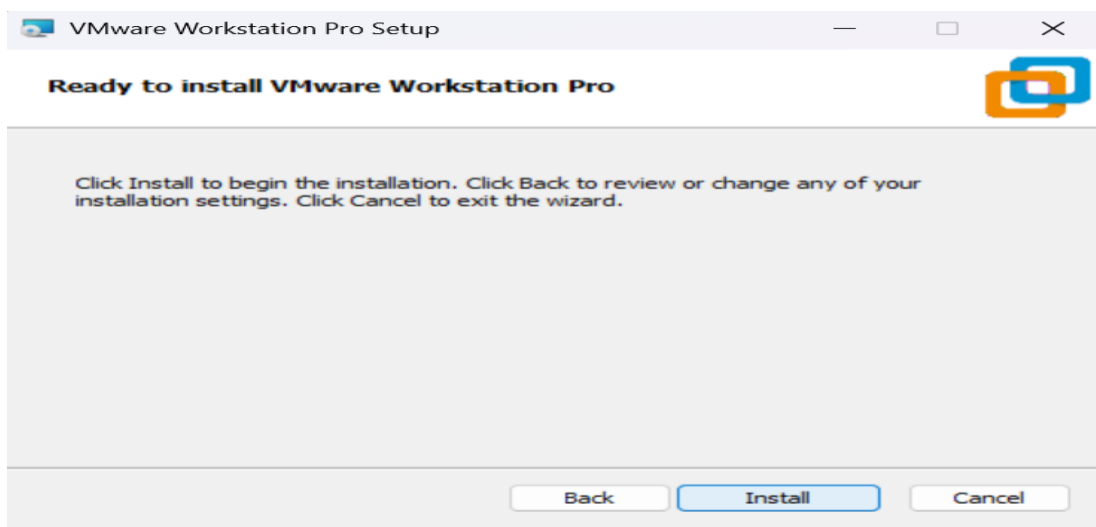
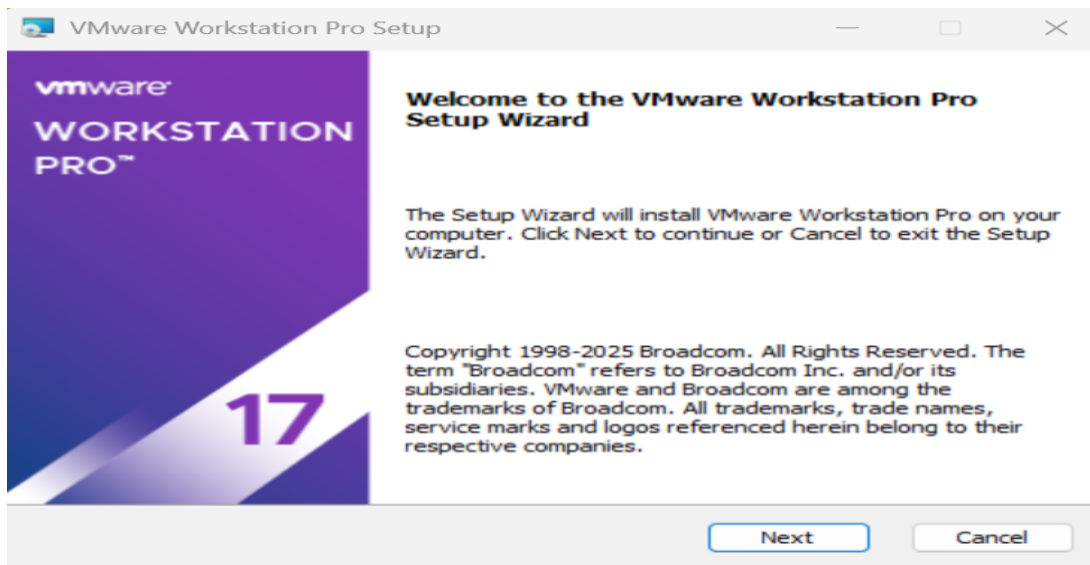
Step 1: type in “vmware” in the search bar and click techspot as shown below

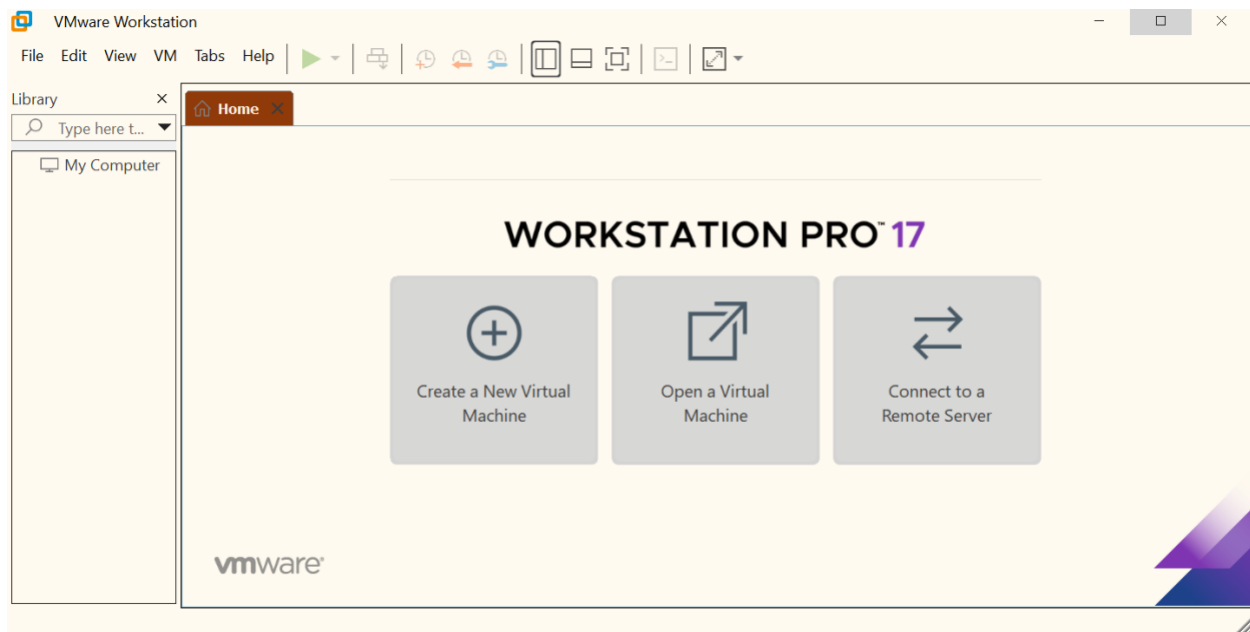


Step 2: Download vmware

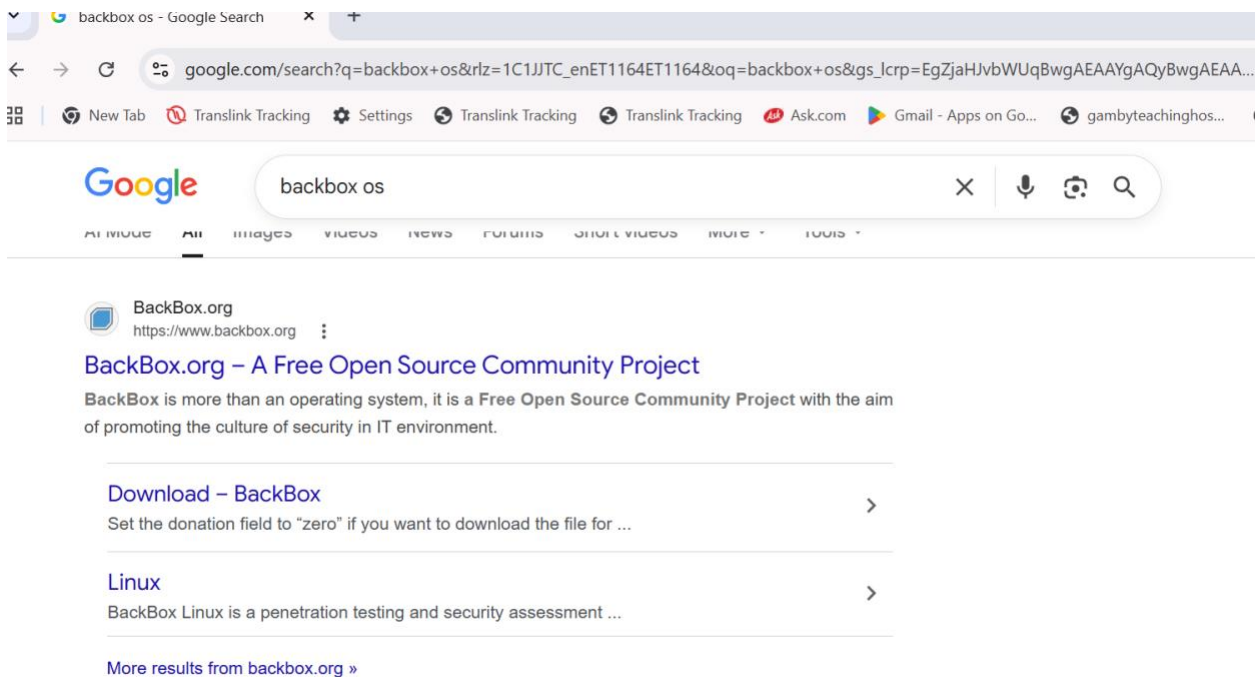


Step 3: follow the installation process as shown below

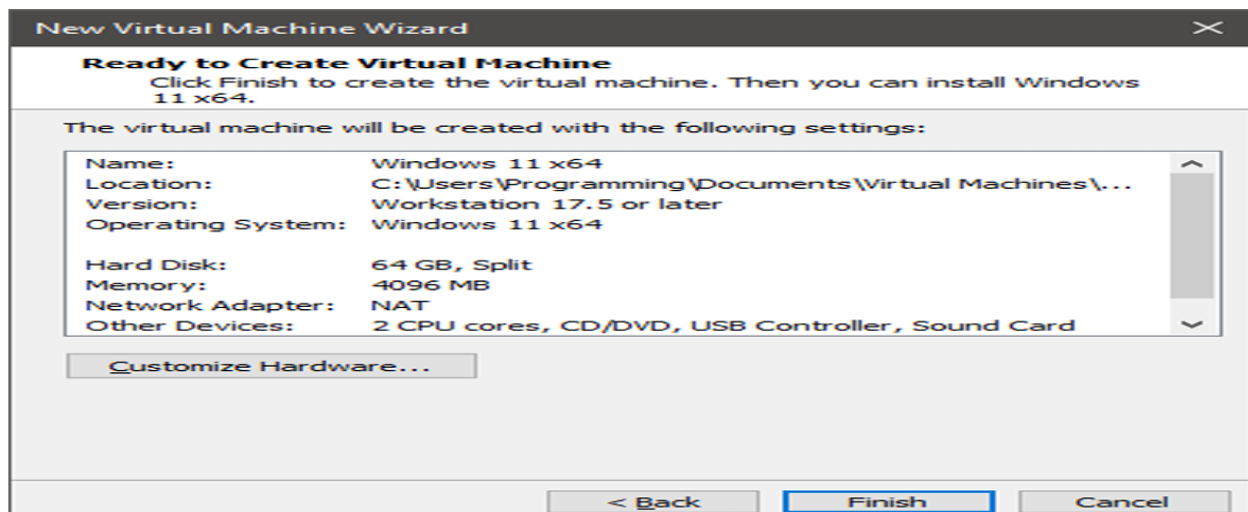
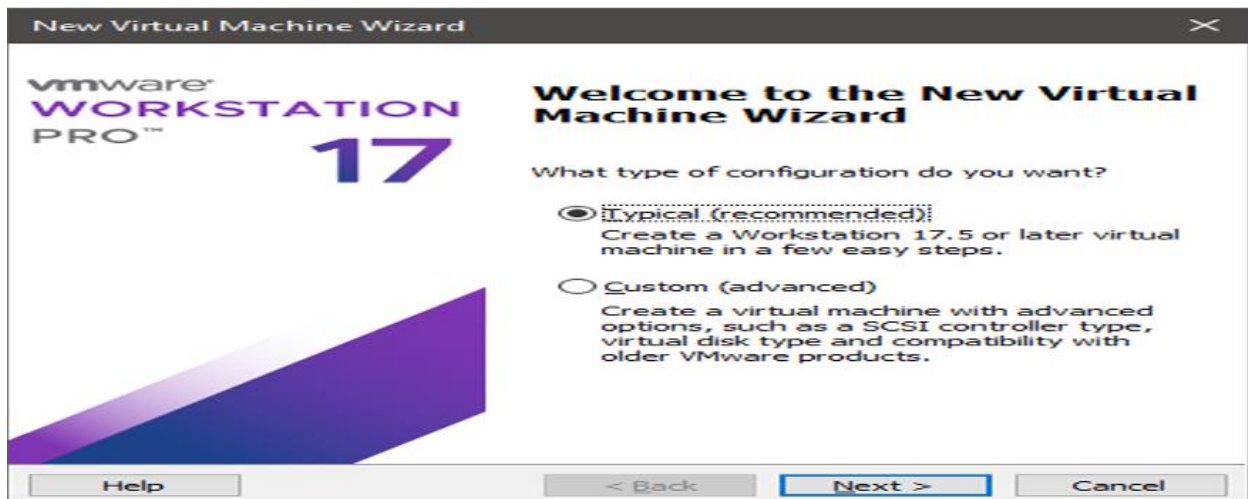




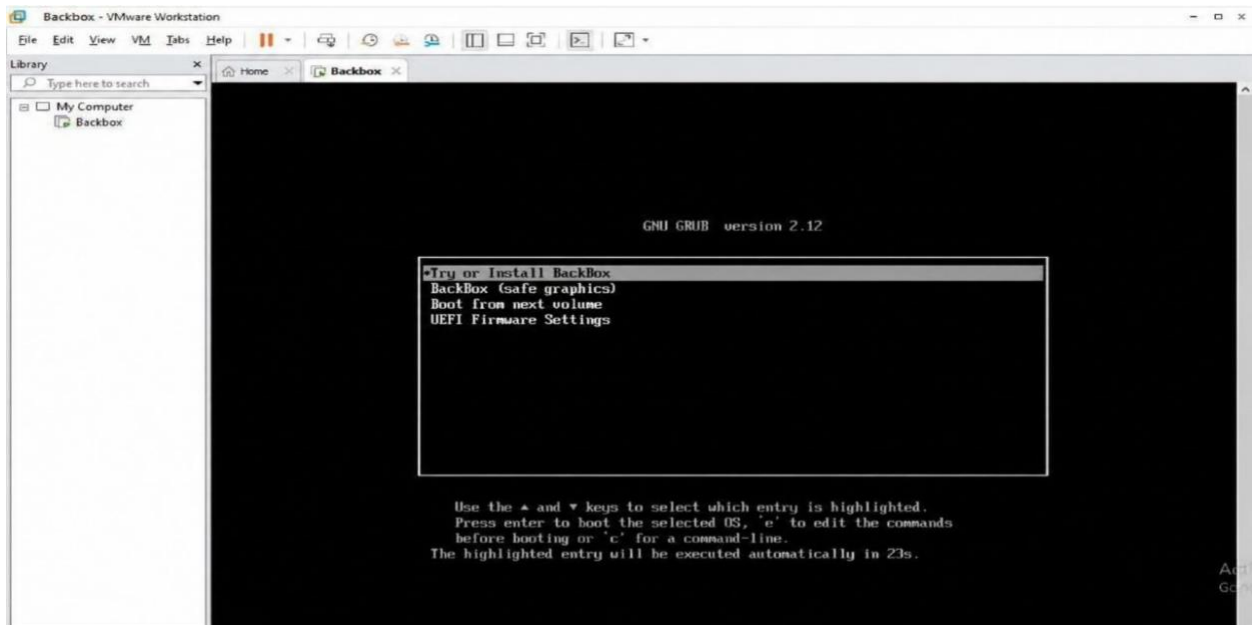
Step 4: we will type in “backbox os” and download an ISO image file from the site.



Step5: Install Backbox linux on VMwareworkstation

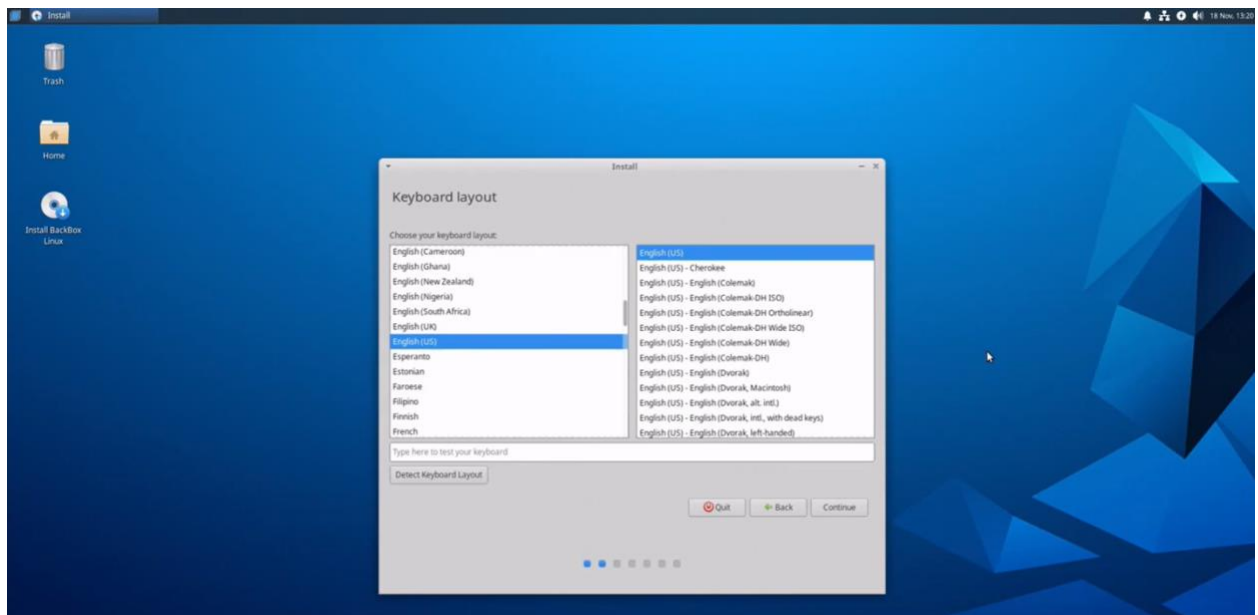


Step 6: click “Try or install Backbox”

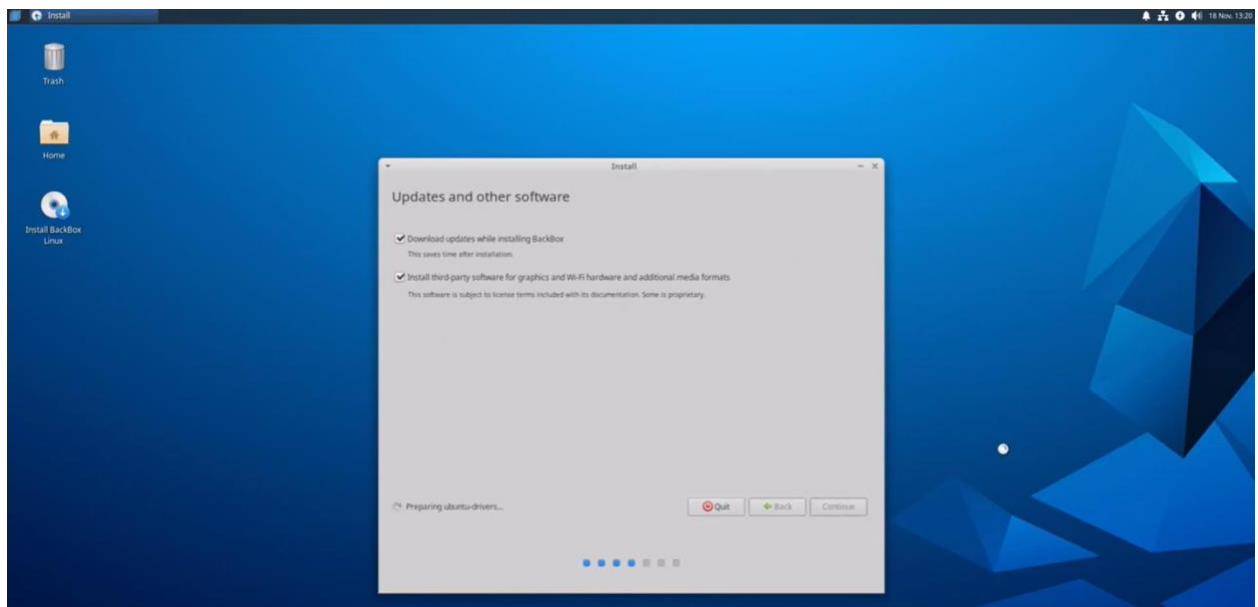


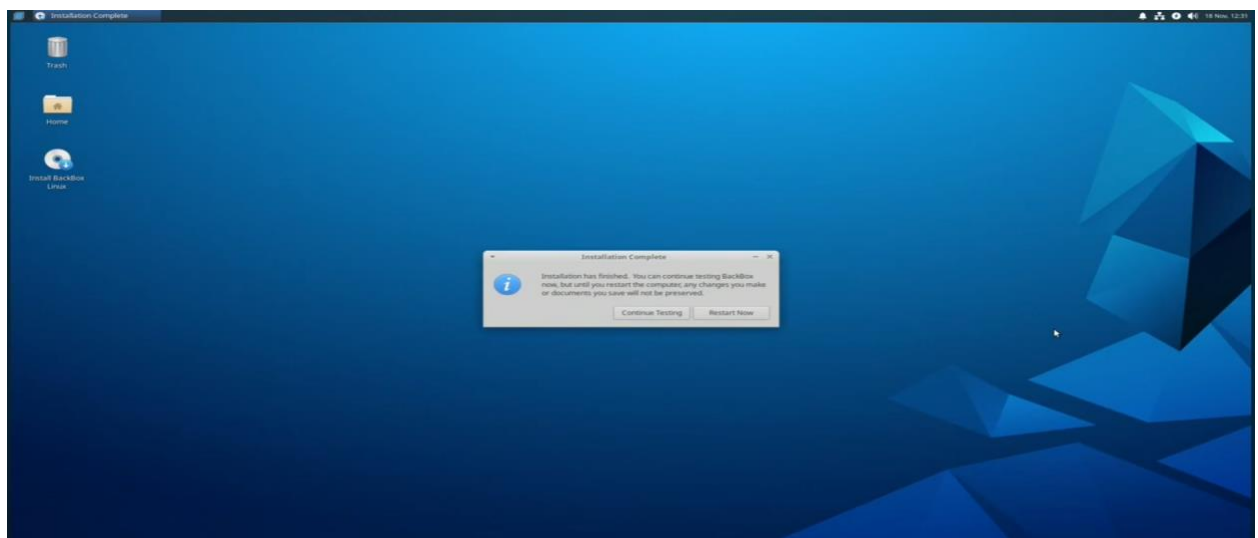
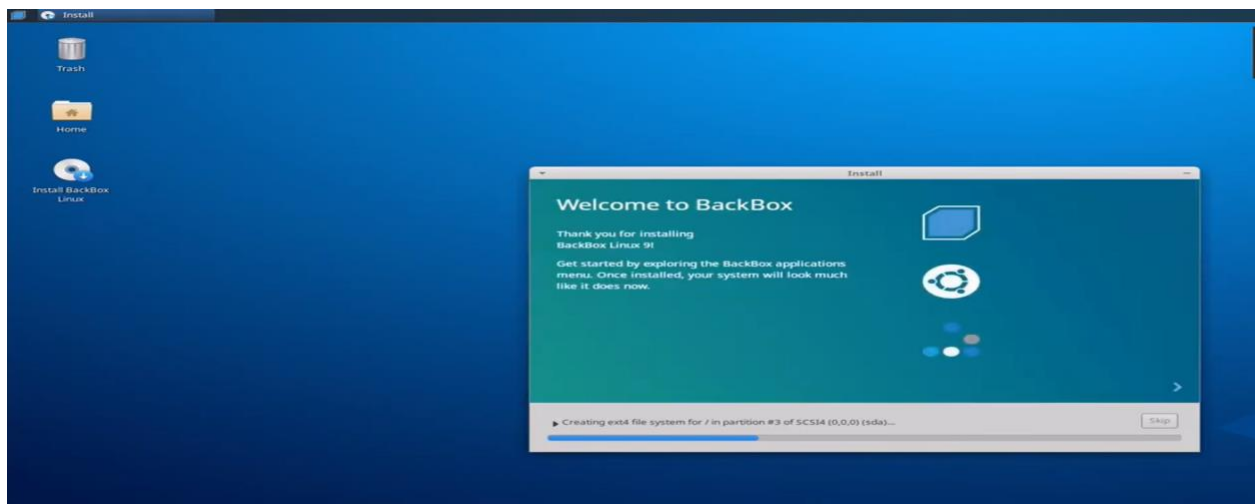
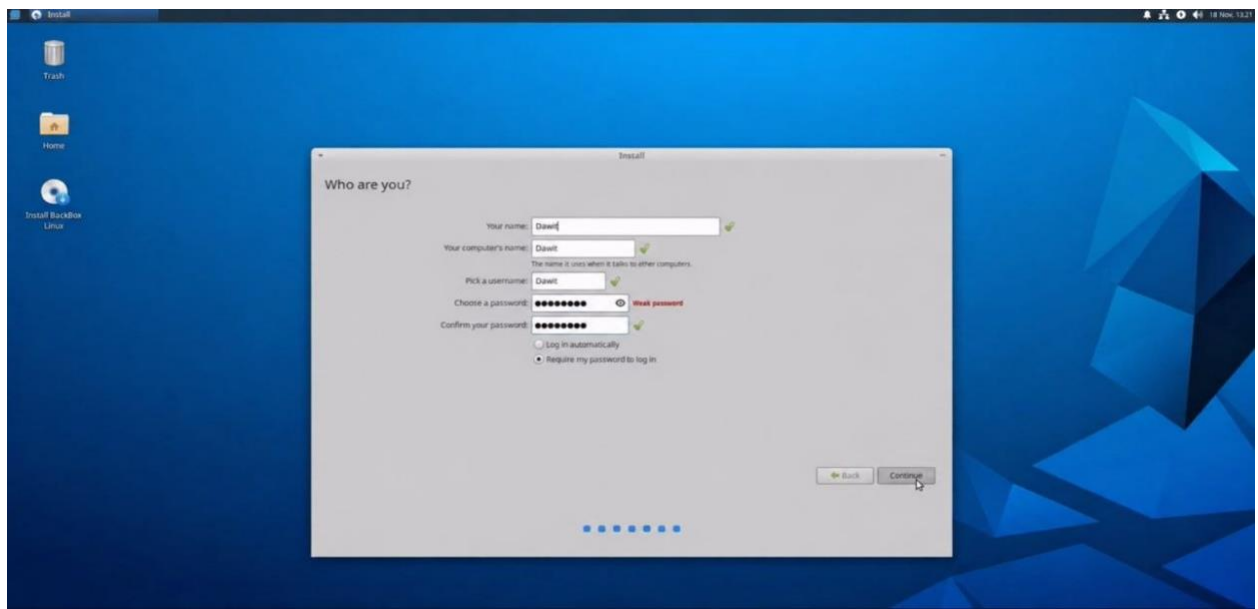
Step7: Then we follow the installation process as shown below





Click Continue:





Installation complete.

Issues (Problems Faced)

Slow performance

- System lagging or freezing during installation

No internet connection

- Cannot update or install packages

File loss during installation

Solutions

- Switch to another device to improve the performance
- Switch to another network to get better connection

Advantages

- Lightweight and fast Linux distribution
- Pre-installed security and penetration testing tools
- Easy installation on virtual machines
- Based on Ubuntu → strong community support
- Good hardware compatibility

Disadvantages

- Requires good system resources for smooth VM performance
- Some tools require advanced knowledge
- Limited support compared to bigger distros like Kali Linux
- Graphics issues may occur in VM
- Not ideal for beginners in cybersecurity

Conclusion

Installing **BackBox Linux** on **VMware Workstation** is a practical way to explore a penetration testing environment without affecting the main system. Although users may face issues like boot

errors, slow performance, or network problems, these can be resolved with proper configuration and resource allocation. The use of the **ext4 filesystem** ensures stability and efficiency. Overall, BackBox provides a lightweight and secure platform, especially useful for learning cybersecurity in a safe virtual environment.

Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a technology that allows you to create virtual (not physical) computers inside one real computer. In simple words, it lets one physical machine act like many computers at the same time.

This means a single computer, called the **host**, can run multiple virtual machines (VMs). Each virtual machine works like a real computer. It has its own operating system (OS), storage, memory, and settings.

For example, you can have one computer running Windows, and inside it you can create another virtual machine running Linux. Even though they are on the same physical computer, they act like separate systems.

Each virtual machine is independent and isolated, so what happens in one VM does not affect the others or the main system.

Why Use Virtualization?

Virtualization is very useful because it helps use computer resources better and gives more flexibility.

Instead of using many physical computers, you can use one machine to run several virtual machines. This saves hardware, electricity, and space. For example, instead of buying two computers, you can run both Windows and Linux on one system using virtual machines.

It also reduces cost because fewer physical machines are needed.

Another important advantage is **safety (isolation)**. If you are doing risky tasks like testing software, hacking tools, or unknown programs, you can do it inside a virtual machine. If something goes wrong, your main computer will not be affected.

Virtualization also makes work faster and easier. You can quickly create new virtual machines, copy them, or delete them. It also allows features like:

- Taking snapshots (saving the current state)
- Easy backup and recovery
- Testing new systems without risk

Because of this, many companies use virtualization to create test environments or run servers without buying new hardware.

How Does Virtualization Work?

Virtualization works using a special software called a **hypervisor** (also known as a virtual machine monitor).

The hypervisor controls and manages the virtual machines. It takes the physical resources of the computer, like CPU, RAM, and storage, and shares them between different virtual machines.

There are two main types of hypervisors:

Type 1 Hypervisor

This type runs directly on the physical hardware without needing another operating system.

Examples include:

- VMware ESXi
- Microsoft Hyper-V
- KVM

These are usually used in big systems like servers and data centers.

Type 2 Hypervisor

This type runs on top of an existing operating system, just like a normal application.

Examples include:

- VMware Workstation
- Oracle VirtualBox

These are commonly used by students and individuals on personal computers.

When a virtual machine is running, any request it makes (like using the CPU or saving a file) goes through the hypervisor. The hypervisor then manages and gives the required resources from the real computer.

Conclusion

In conclusion, virtualization allows one physical computer to run many virtual machines at the same time. It helps improve efficiency, reduce cost, and increase flexibility. It also provides a safe environment for testing and learning.

Virtualization is widely used in modern technology, including data centers, cloud computing, and software development.

In our case, using tools like **VMware Workstation** or **Oracle VirtualBox** allows us to install and run **BackBox Linux** easily without needing a separate physical computer. This shows how virtualization makes learning and experimentation simple and safe.

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